Traffic Collision Data Analysis Report

California SWITRS Dataset | PySpark + Python

Apache PySpark 3.5 | Pandas | Matplotlib / Seaborn | GeoPandas

Distributed ETL | EDA | Spatial Analysis | Time-Series

Statewide Integrated Traffic Records System (SWITRS) — California Highway Patrol

SECTION 1

Introduction

This project analyses large-scale California traffic collision data using **Apache PySpark** to uncover spatial, temporal, and environmental patterns affecting road safety. Due to the large size of the dataset, PySpark was used for scalable distributed processing, while Pandas and visualisation libraries were used for exploratory analysis on sampled data.

Why PySpark?	The SWITRS dataset spans hundreds of thousands of records across four relational tables. Standard Pandas would be memory-constrained. PySpark enables distributed in-memory computation, lazy evaluation, Spark SQL querying, and MLlib integration — all at full dataset scale.

SECTION 2

Data Preparation

The dataset was loaded from CSV/SQLite files into PySpark DataFrames. Schema inspection ensured correct data types and structure. Sampling (5%, seed=42) was applied **only during visualisation** to maintain system stability while preserving representative trends. All aggregations and ETL queries ran on the full dataset.

Dataset Structure

Table	Description	Key Attributes	Volume
collisions	Core incident records	case_id, date, severity, weather, lighting, county	~600K+ rows
parties	Drivers, passengers, pedestrians	party_type, party_age, violation_code	~1.2M+ rows
victims	Injury detail per person	victim_age, degree_of_injury	~800K+ rows
locations	Geographic coordinates	latitude, longitude, road_type	~600K+ rows

SECTION 3

Data Cleaning

Missing values were examined column-wise across all tables. A **rule-based approach** was used instead of statistical imputation. Numeric count fields (killed_victims, injured_victims) were filled with zero, categorical variables were filled with 'Unknown', and records with missing critical attributes were removed. Duplicate records were eliminated. Outliers were identified using the **Interquartile Range (IQR) method** but retained to preserve rare but severe events.

Column Type	Strategy	Rationale
Numeric count fields (killed_victims, injured_victims)	**Fill with 0**	Missing count = no recorded victims, not an unknown quantity

Categorical fields (weather_1, lighting, road_surface)	**Fill with 'Unknown'**	Preserves record while flagging missing context
Critical identifiers (case_id, collision_date, severity)	**Drop record**	Cannot analyse an incident with no identifying information
Sparse columns (>50% missing)	**Column removed**	Near-empty columns add noise without analytical value

SECTION 4

Exploratory Data Analysis

EDA was conducted using three complementary techniques: **univariate** analysis for single-variable distributions, **segmented univariate** for group-wise patterns, and **bivariate** for relationship detection. Visualisations were generated using Pandas/Matplotlib after sampling from Spark DataFrames. All 13 charts are reproduced below with their analytical findings.

4.1 Collision Severity Distribution

Visualisation: Bar chart | Analysis type: Univariate

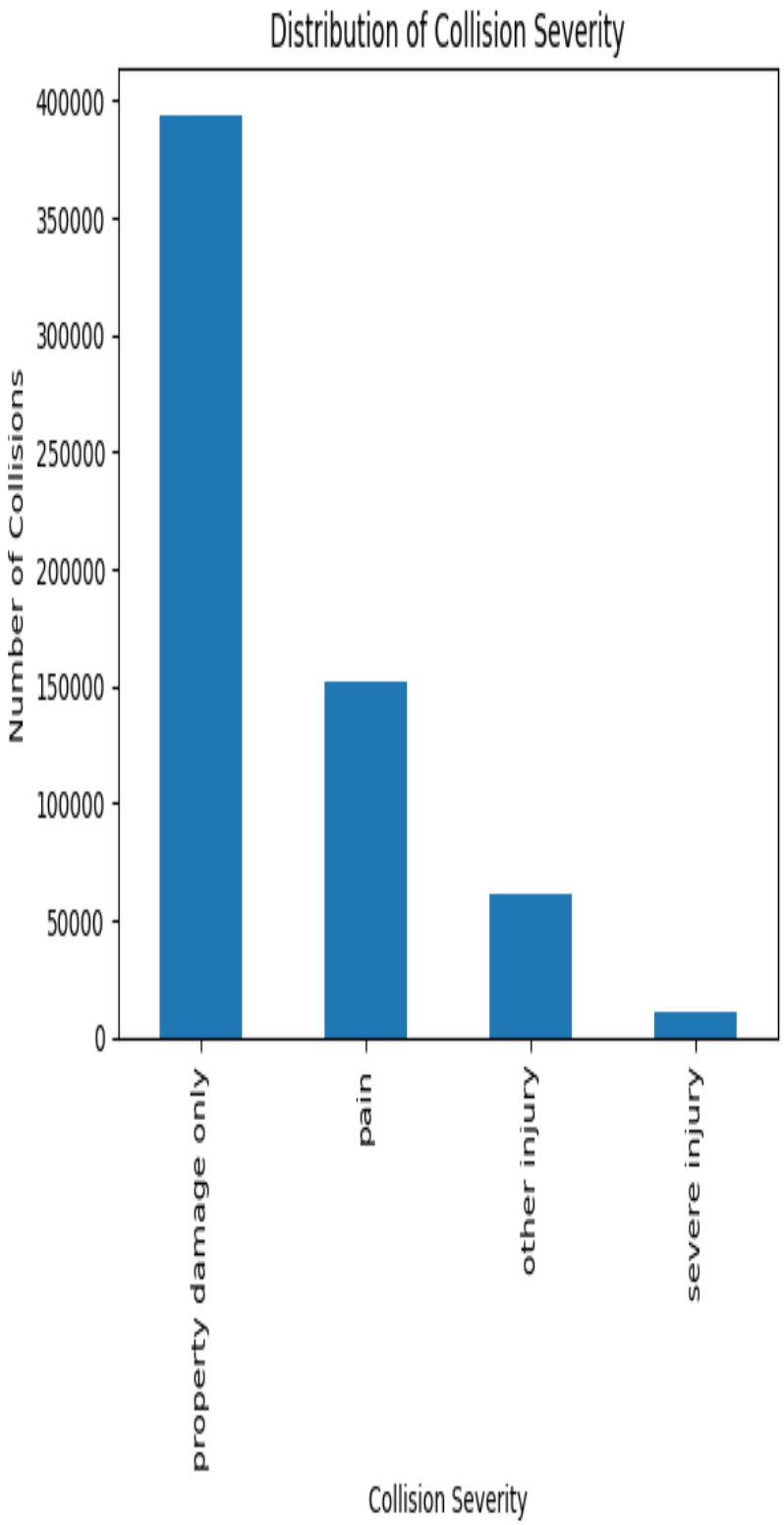


Fig 4.1 — Distribution of Collision Severity

Result	Property damage only: ~390,000 (dominant). Pain: ~155,000. Other injury: ~65,000. Severe injury: ~18,000. Fatal: <5,000.
Insight	High-frequency low-severity crashes dominate overall statistics but do not reflect true societal impact. Fatal collisions are rare but irreversible.

Outcome	Road safety policies should prioritise reducing high-severity and fatal collisions rather than optimising for total count reduction alone.

4.2 Weather Conditions During Collisions

Visualisation: Bar chart | Analysis type: Segmented Univariate

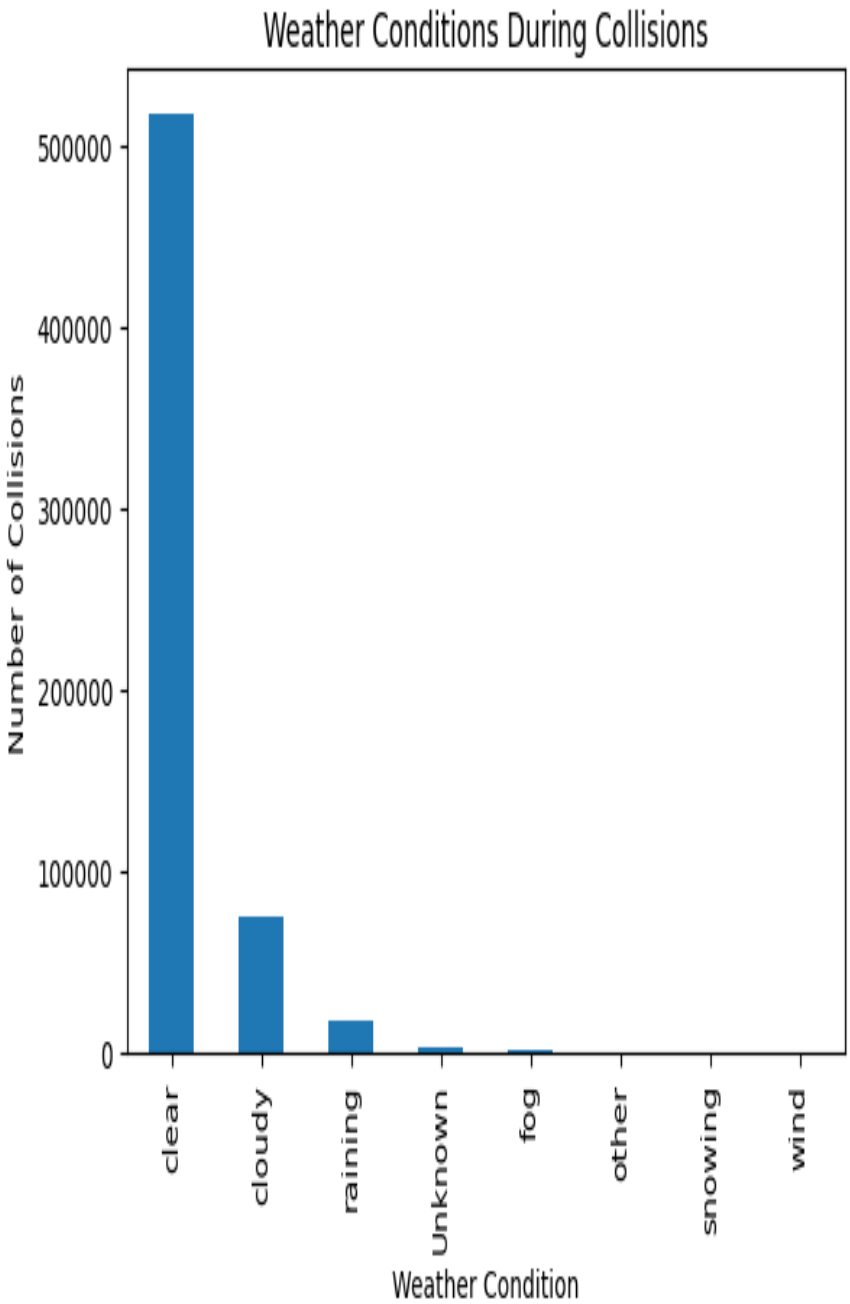


Fig 4.2 — Weather Conditions During Collisions

Result	Clear weather: ~500,000 collisions (majority). Cloudy: ~80,000. Raining: ~35,000. Fog/Snow/Wind: <10,000 each.

Insight	Traffic exposure and driving volume influence collisions far more than adverse weather. Most driving happens in clear conditions.
Outcome	Road safety planning should focus on normal driving conditions — not only adverse weather periods.

4.3 Victim Age Distribution

Visualisation: Histogram | Analysis type: Univariate

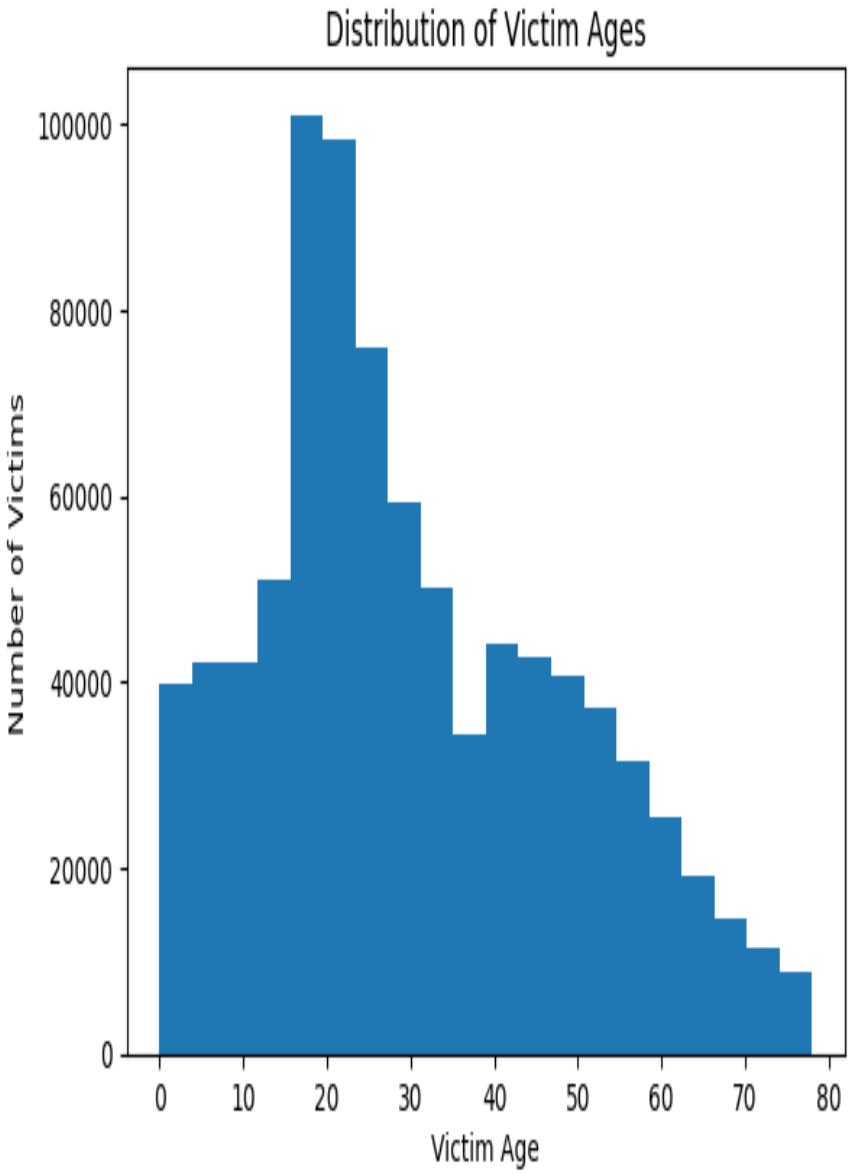


Fig 4.3 — Distribution of Victim Ages

Result	Peak concentration in the 20–40 age group (~80,000–100,000). Very young (<10) and elderly (70+) are less frequent.
Insight	The active working population faces higher road risk due to frequent commute-related travel and higher driving exposure.
Outcome	Targeted awareness campaigns should focus on commuters and working-age drivers (18–40).

4.4 Collision Severity vs Number of Victims

Visualisation: Box plot | Analysis type: Bivariate

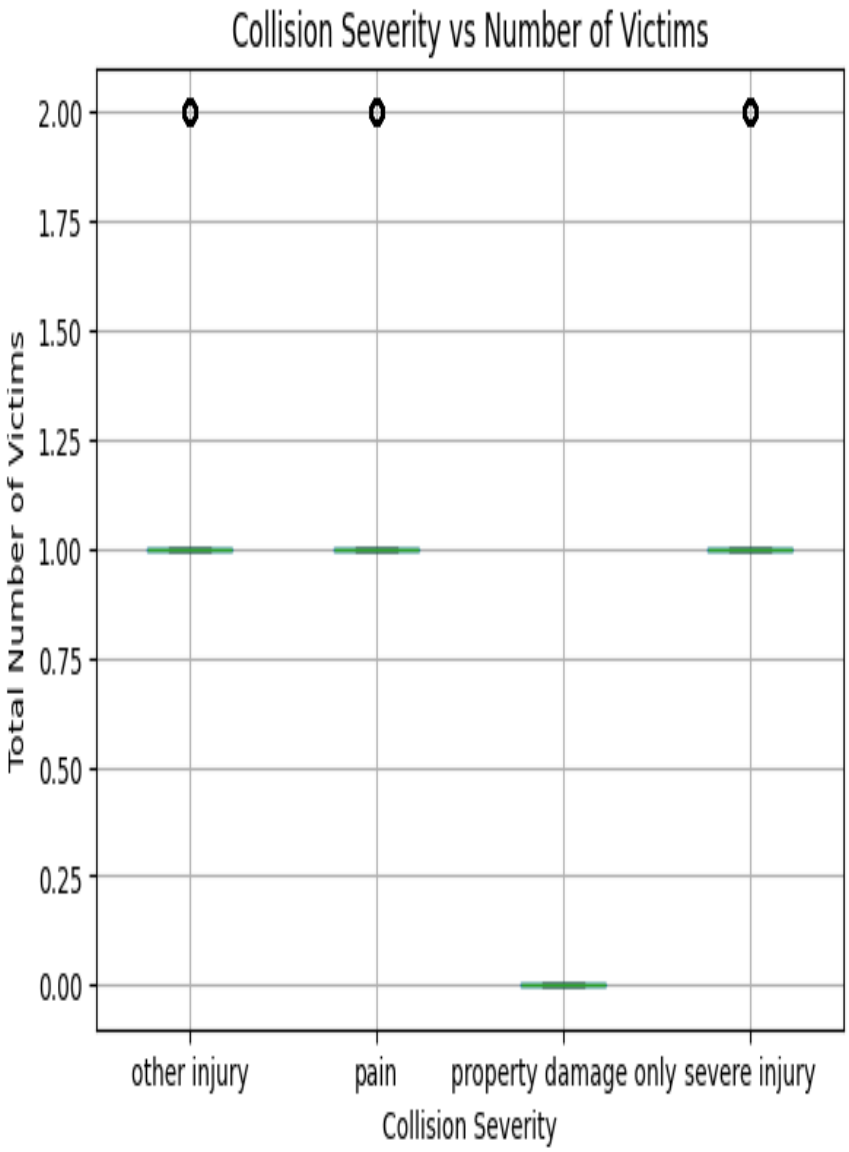


Fig 4.4 — Collision Severity vs Total Number of Victims

Result	Higher severity collisions show higher median victim counts and wider spread. Severe injury and fatal collisions have the broadest distributions with highest outliers.
Insight	Multi-victim crashes escalate injury severity and emergency response burden simultaneously. Higher severity = more victims, compounding the societal cost.
Outcome	Emergency dispatch algorithms should factor estimated victim count — not just incident type — to improve triage efficiency.

4.5 Weather Conditions vs Collision Severity

Visualisation: Grouped bar chart | Analysis type: Bivariate

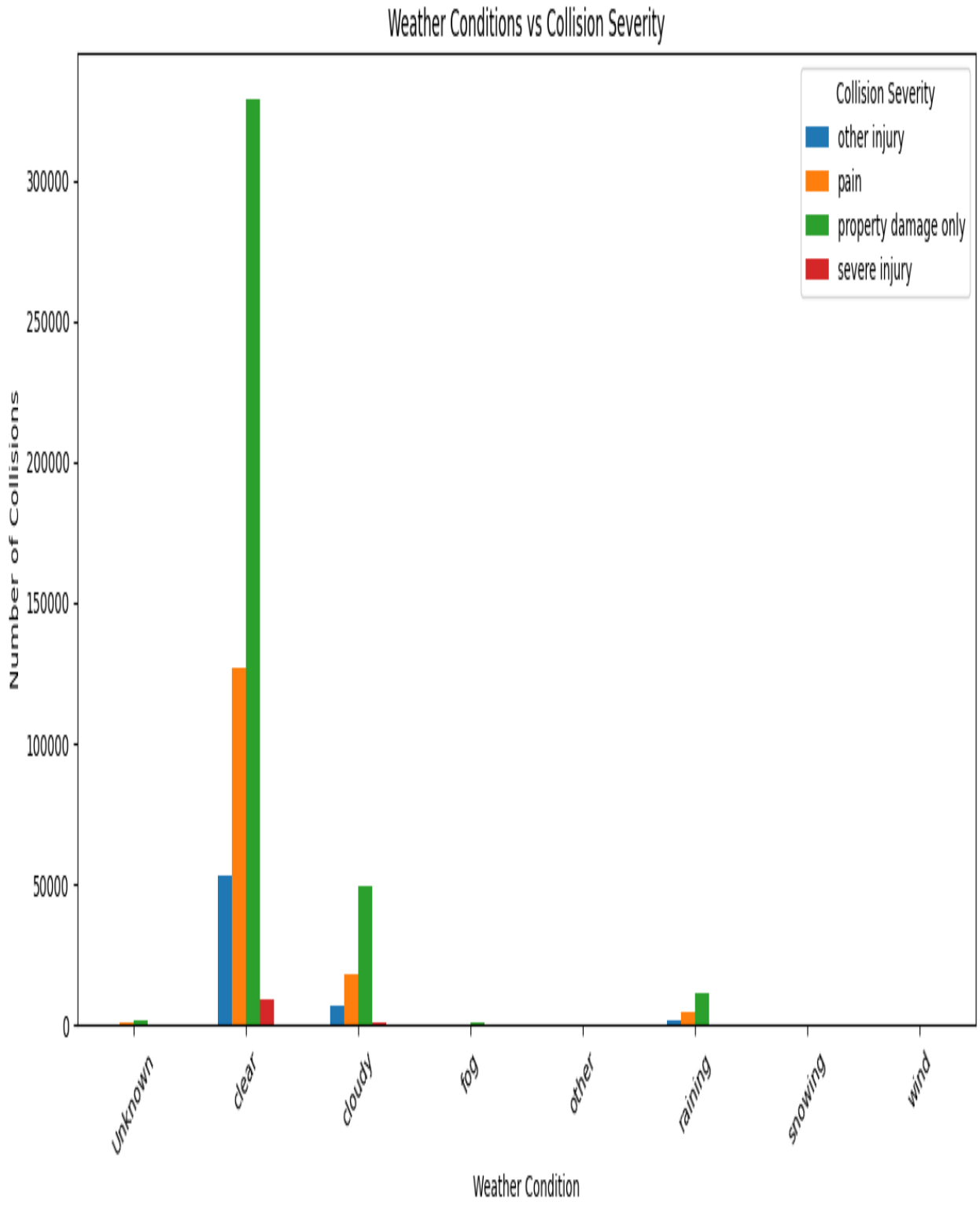


Fig 4.5 — Weather Conditions vs Collision Severity

Result	Clear weather produces the highest absolute count of severe collisions at every severity level. Adverse weather contributes a small share despite conventional assumptions.

Insight	Higher traffic volume and speed during clear weather increase severity despite favourable conditions. This is counterintuitive but data-supported.
Outcome	Road safety interventions should target normal weather conditions through speed control and awareness programmes.

4.6 Lighting Conditions vs Collision Severity

Visualisation: Heatmap (Seaborn) | Analysis type: Bivariate Categorical

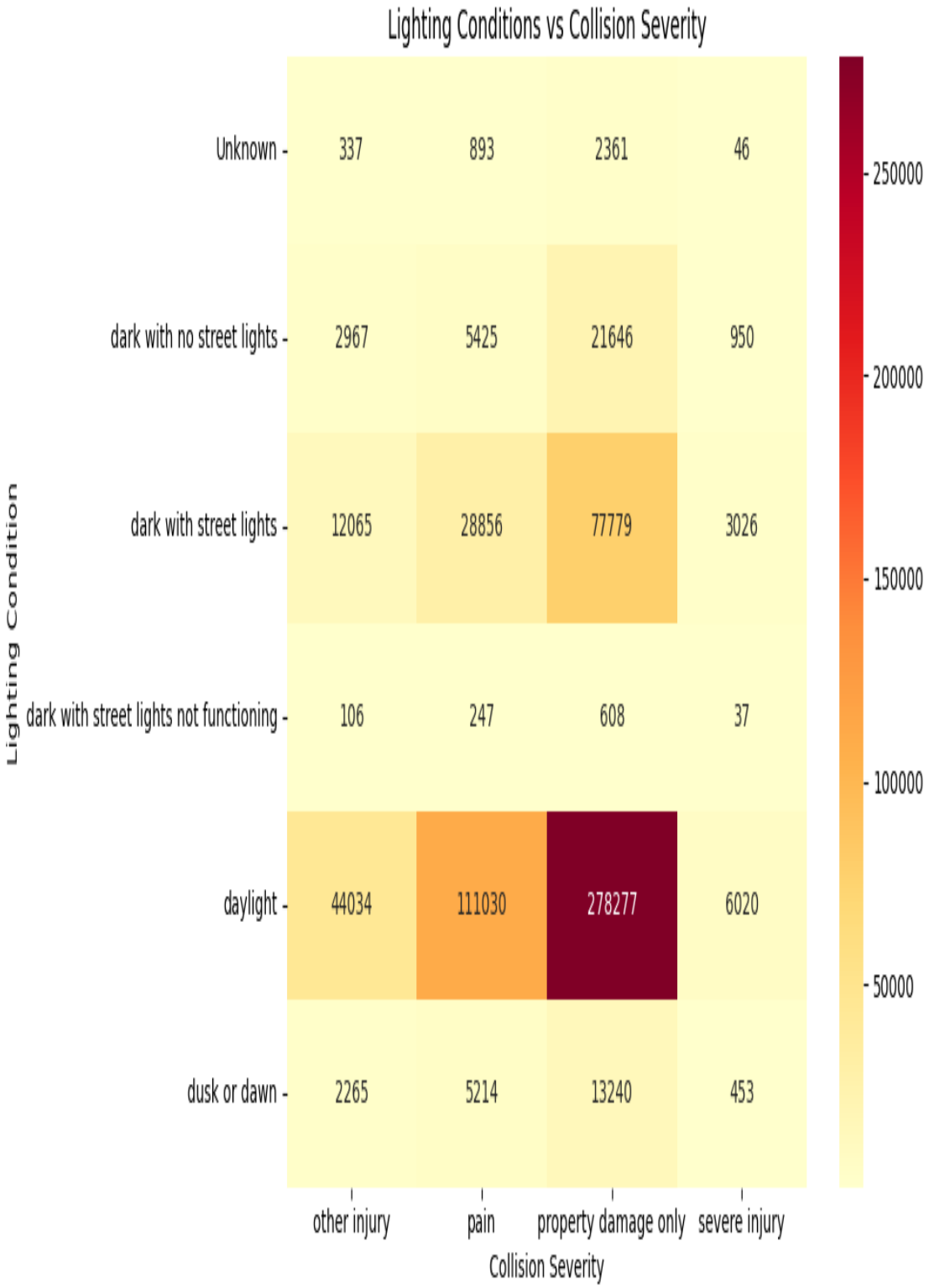


Fig 4.6 — Lighting Conditions vs Collision Severity (Heatmap)

Result	Daylight: 44,034 / 111,030 / 278,277 / 6,020 across severity levels. Dark (no street lights): 2,967 / 5,425 / 21,646 / 950. The severe injury rate is proportionally higher in unlit conditions.
Insight	Poor or no lighting correlates with higher severity rates per incident. Reduced visibility significantly increases serious injury risk.

Outcome	Targeted street lighting upgrades in high-collision, unlit zones would reduce severe collision rates with a contained infrastructure investment.

4.7 Weekday-Wise Collision Trends

Visualisation: Line chart | Analysis type: Segmented Time-Series

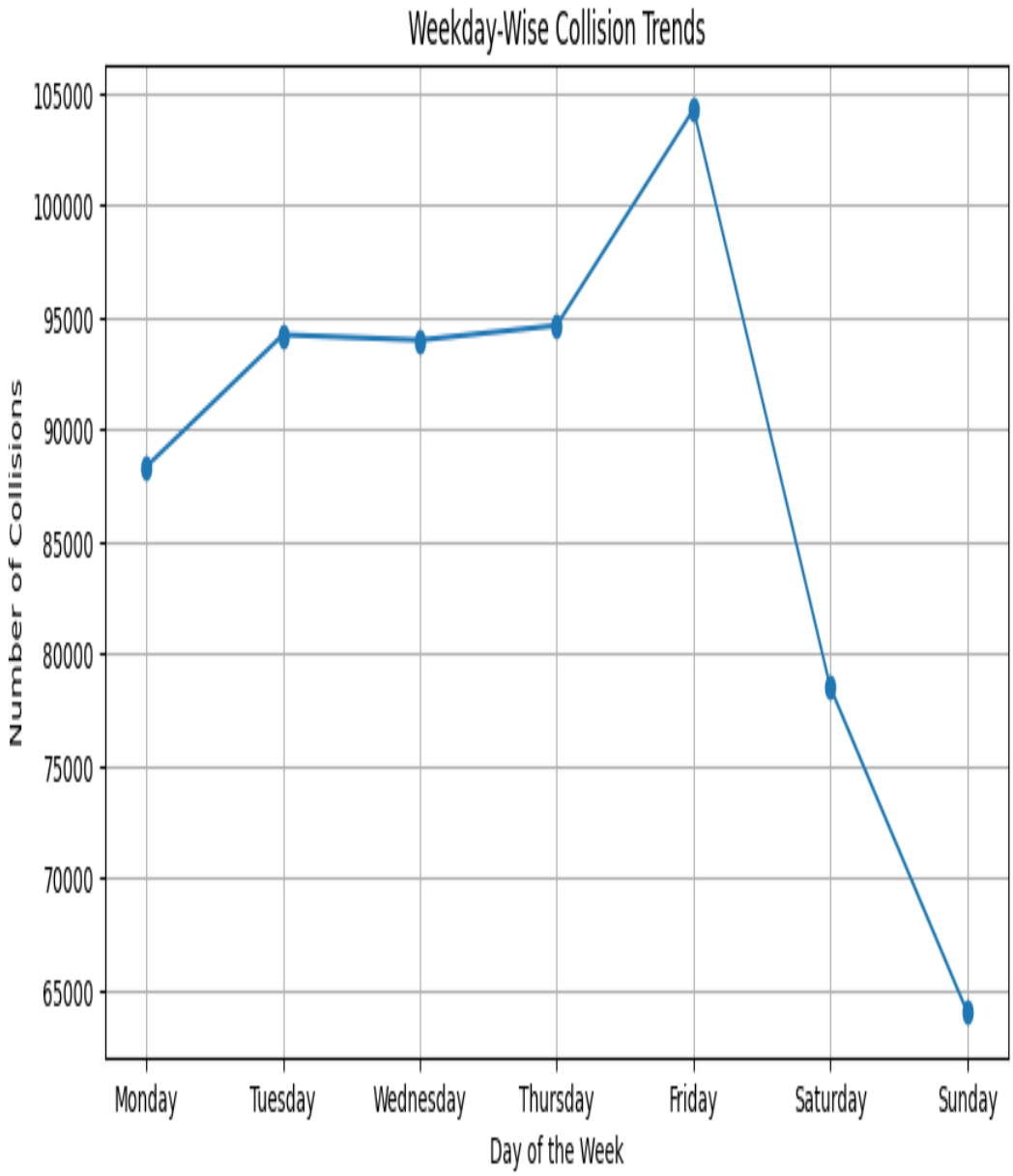


Fig 4.7 — Weekday-Wise Collision Trends

Result	Friday peak: ~104,000 collisions. Mid-week stable: ~95,000. Monday: ~89,000. Sunday lowest: ~65,000.
Insight	Congestion, end-of-week fatigue, and regular commuting patterns contribute to increased weekday collision risk, peaking on Fridays.
Outcome	Weekday-focused enforcement — especially Friday 15:00–18:00 — and signal optimisation can deliver the highest marginal safety improvement.

4.8 Spatial Distribution of Collisions

Visualisation: County bar chart + Choropleth map (GeoPandas) | Analysis type: Spatial Analysis

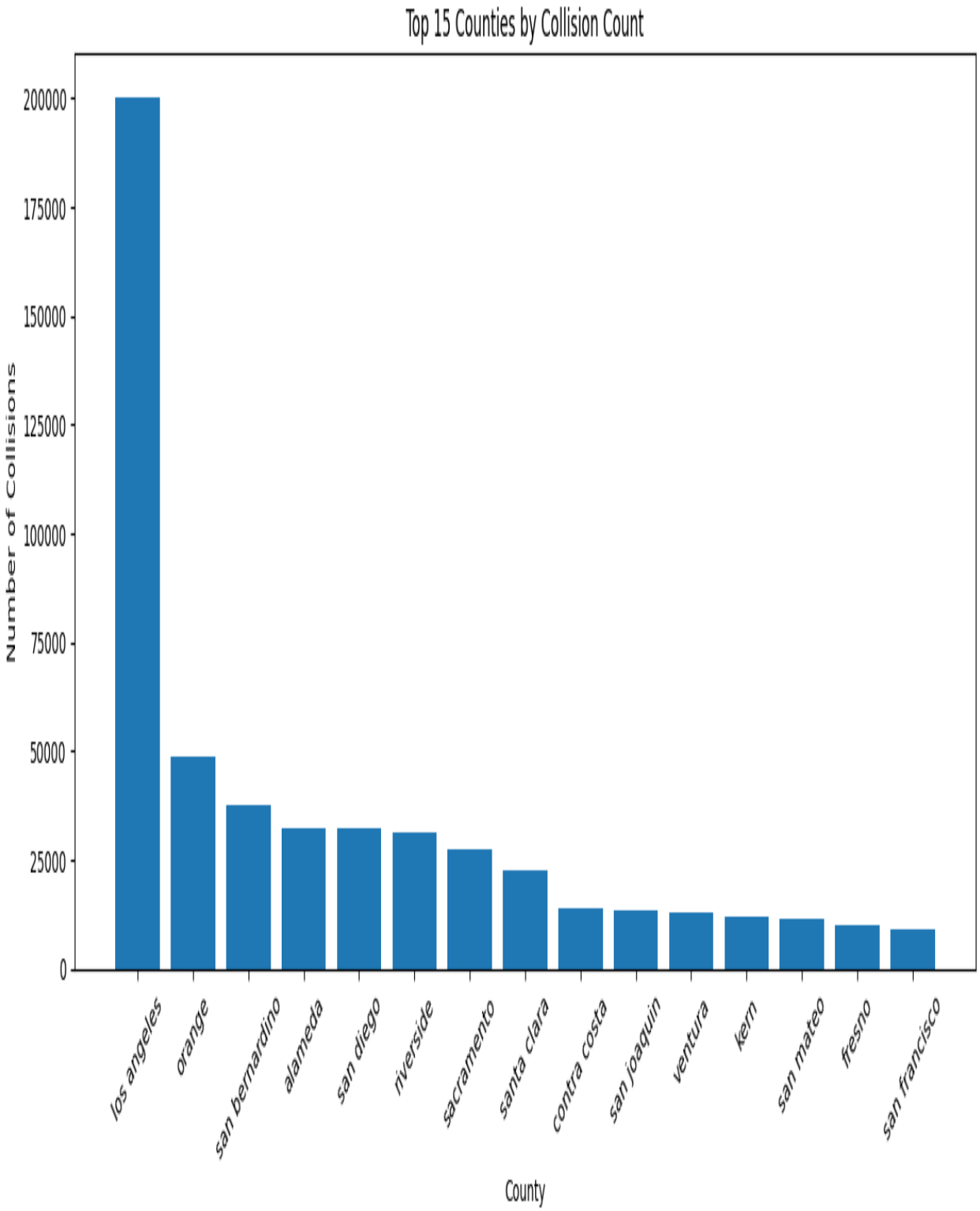


Fig 4.8a — Top 15 Counties by Collision Count

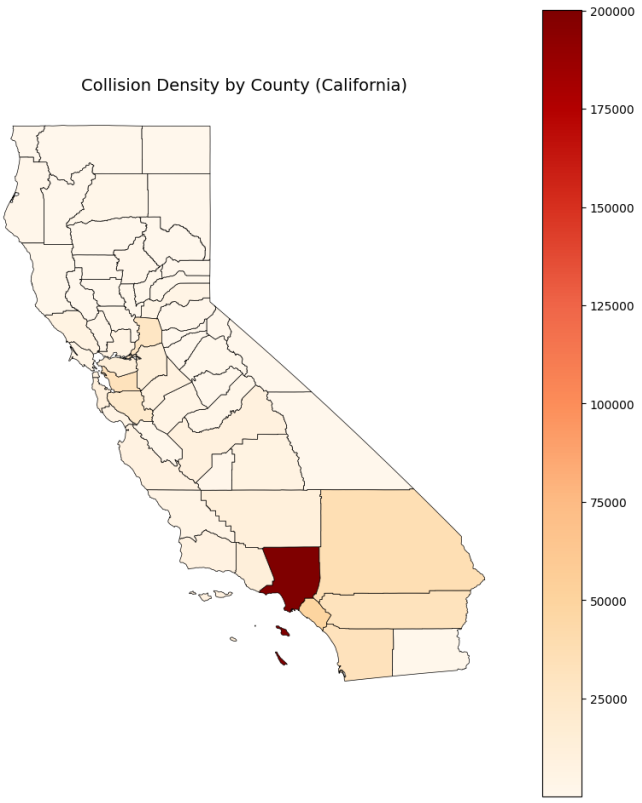


Fig 4.8b — Collision Density Map (California)

Result	Collisions cluster in urban counties (LA, San Diego, Riverside, Orange). A small number of counties account for a disproportionately high share of statewide collisions.
Insight	Urban density, road complexity, and traffic concentration are key contributors. Risk is geographically concentrated — the 80/20 principle applies strongly.
Outcome	Hotspot-based infrastructure investment is more cost-effective than uniform statewide spending. GIS density maps should directly inform speed camera placement and intersection redesign priorities.

4.9 Collision Analysis by Geography

Visualisation: Latitude/Longitude scatter plot | Analysis type: Spatial Analysis

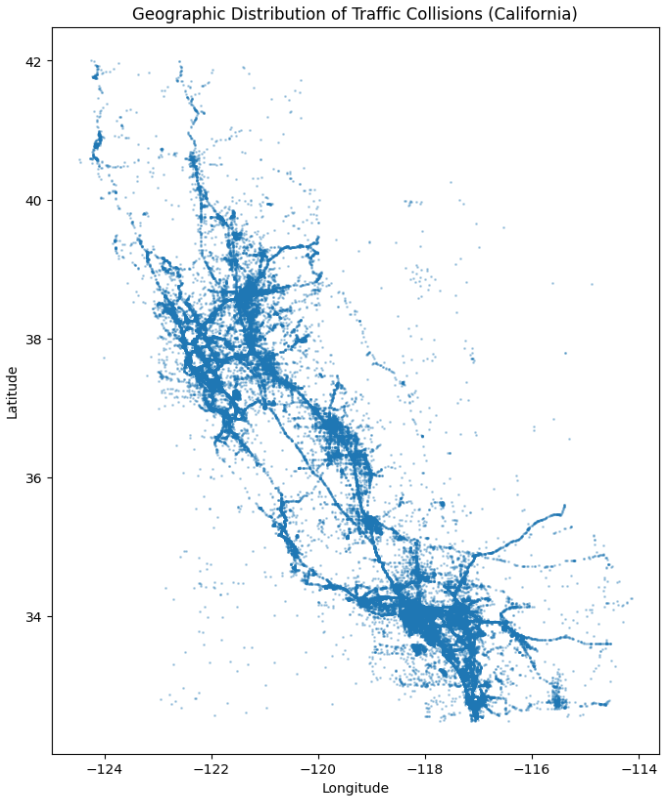
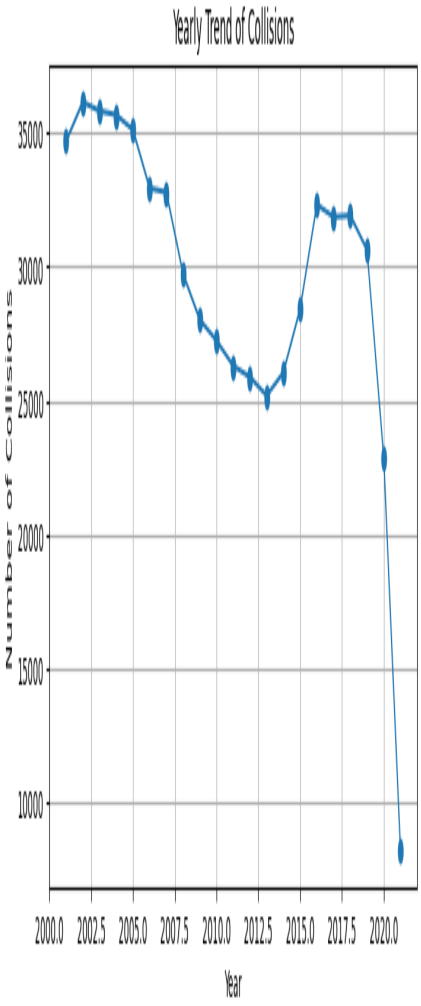


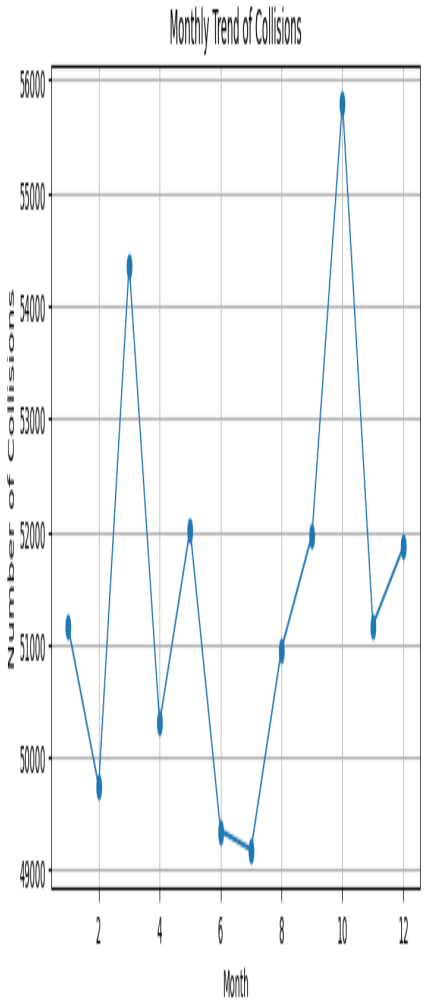
Fig 4.9 — Geographic Distribution of Traffic Collisions (California)

Result	Collisions are densely clustered along urban corridors and major highways (I-5, US-101, I-405). Rural areas show sparse, near-zero occurrence. Dense city clusters visible at LA, San Francisco, San Diego.
Insight	High traffic density and urban road complexity significantly increase collision risk. Collision geography precisely traces California's major transportation infrastructure.
Outcome	Urban collision hotspots should be prioritised for infrastructure upgrades, enforcement, and safety interventions. The scatter map can directly drive patrol route design.

4.10 Collision Trends Over Time

Visualisation: Line charts (yearly, monthly, hourly) | Analysis type: Segmented Time-Series





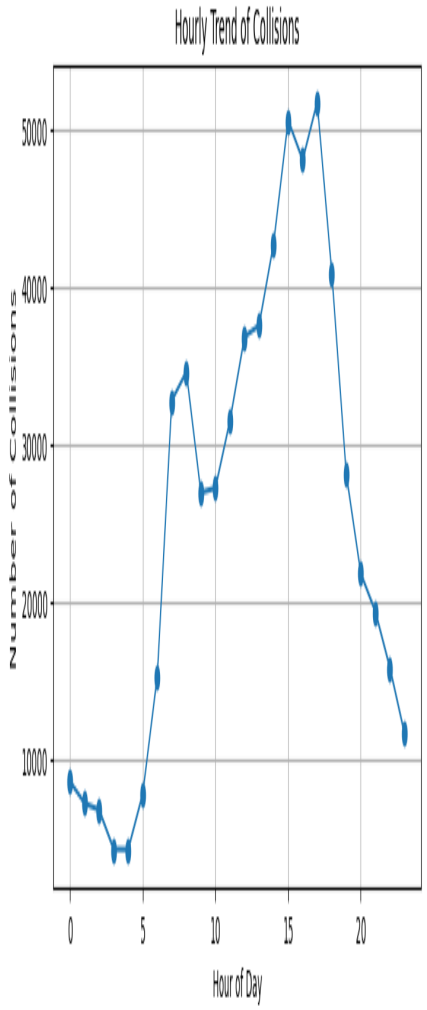


Fig 4.10a — Yearly Trend | Fig 4.10b — Monthly Trend | Fig 4.10c — Hourly Trend

Scale	Key Finding	Insight	Action
Yearly	Peak ~2001–05 (33–35K/yr). 2020 sharp drop to ~8K.	COVID drop is a natural experiment: volume reduction is the most powerful safety tool.	Plan for traffic demand growth and mobility shifts.
Monthly	Range: 49K–56K. October peak. February lowest.	Seasonal travel and holiday mobility drive fluctuations.	Deploy seasonal enforcement — especially October.
Hourly	Peak: 15:00–18:00 (~50K). Morning peak: 07:00–09:00.	Congestion, fatigue and time pressure combine during commute hours.	Time-index enforcement to 15:00–18:00 weekdays.

SECTION 5

ETL Query Results (PySpark SQL)

ETL operations were implemented using **PySpark SQL** on the full registered DataFrame view. Results identify top counties, peak months, common weather conditions, fatal collision percentage, dangerous hours, road surface and lighting conditions contributing to collisions.

Query	Business Question	Key Finding	Action Signal
Q4.1 Top 5 Counties	Which counties need priority investment?	LA, San Diego, Riverside, San Bernardino, Orange dominate statewide counts.	Direct resources to these 5 counties first.
Q4.2 Peak Month	When should seasonal enforcement peak?	October is consistently the highest collision month.	Pre-position resources before October each year.
Q4.3 Weather	Does adverse weather drive frequency?	Clear weather: majority of all collisions. Rain/Fog/Snow: <10% combined.	Target normal-weather, high-volume corridors.
Q4.4 Fatal %	What share of incidents are fatal?	Small percentage — but highest policy priority given irreversible human cost.	Use severity-weighted metrics, not raw counts.
Q4.5 Dangerous Hours	When should peak-hour deployments occur?	15:00–18:00 evening commute is the single most dangerous window.	Time-index enforcement to 15:00–18:00.
Q4.6 Road Surface	Do road conditions drive frequency?	Dry/normal surfaces dominate — exposure effect, not inherent danger.	Maintenance is critical; surface is a risk multiplier.
Q4.7 Lighting	Which lighting conditions are most dangerous?	Daylight: highest volume. Dark/no street lights: highest severity rate per incident.	Prioritise lighting upgrades in unlit high-severity zones.

SECTION 6

Conclusions & Strategic Recommendations

Final Insights Summary

Theme	Key Finding	Policy Signal
Severity	85%+ low-severity; <1% fatal	Severity-weighted metrics, not raw counts
Weather	80%+ in clear weather	Target normal conditions, not adverse weather
Demographics	Ages 20–40 most affected	Commuter-focused safety campaigns
Time of day	Peak: 15:00–18:00 evening commute	Time-indexed enforcement deployment
Day of week	Friday highest (~104K); Sunday lowest (~65K)	Weekday enforcement priority
Lighting	Dark/unlit amplifies severity rates	Targeted street lighting upgrades
Geography	Urban counties dominate; I-5/US-101/I-405 densest	Hotspot-based investment
Long-term	2020 COVID drop confirms volume = primary lever	Mobility management is most powerful tool

(a) Road Safety — Infrastructure & Enforcement	Identify high-risk locations and peak accident times to guide infrastructure improvements. Prioritise high-collision counties and urban corridors. Deploy targeted enforcement during peak accident hours (morning and evening commutes). Improve intersection design, lane markings, and signage in identified high-risk zones. Use historical collision data to rank locations by risk rather than spreading resources uniformly.
(b) Traffic Management Optimisation	Adjust traffic signal timing during peak weekday hours to reduce congestion-related collisions. Improve road lighting in high-severity zones, especially where night-time crashes are frequent. Incorporate seasonal and monthly trends into traffic planning. Design roads assuming collisions occur mostly under normal driving conditions, not just adverse weather.
(c) Pedestrian and Cyclist Safety	Expand dedicated pedestrian crossings and protected bicycle lanes in dense urban areas. Enforce lower speed limits in zones with high pedestrian and cyclist collision density. Improve visibility measures near schools, markets, and transit hubs. Launch behavioural awareness campaigns targeting both drivers and vulnerable road users.

(d) High-Risk Zone Identification	Use county-level and latitude–longitude density analysis to identify collision hotspots. Classify zones into high, medium, and low risk for proactive intervention. Focus on areas with persistent historical patterns, not just recent spikes. Support local authorities with data-backed hotspot maps for decision-making.
(e) Environmental Factors	Weather alone does not drive collision frequency — traffic volume is the dominant factor. Poor lighting is strongly associated with higher collision severity. Road surface conditions reflect exposure rather than inherent danger, but maintenance is critical. Environmental factors should be treated as risk multipliers, not sole causes.
(f) Predictive Modelling and Proactive Safety	Develop predictive models using: time (hour, weekday, month), location (county, coordinates), environmental conditions (lighting, weather), and collision severity history. Use models to anticipate hotspots before incidents occur. Integrate predictions into policing schedules and infrastructure planning. Enable preventive strategies rather than reactive responses.

Final Insight

The analysis demonstrates strong spatial and temporal clustering of traffic collisions in California. The central conclusion is that **traffic volume, not environmental conditions, is the dominant driver of collision frequency**. The 2020 COVID lockdown is a natural experiment confirming this: when volume dropped, collisions dropped proportionally — more than any enforcement or infrastructure change could achieve in that timeframe. Recommendations include targeted enforcement during peak hours, infrastructure improvements in high-risk zones, enhanced pedestrian and cyclist safety measures, and predictive modelling for proactive intervention. These insights support data-driven traffic safety policy and urban planning.